

Course guide

Vetenskaplig metod och kommunikation (NA310G; 7.5 högskolepoäng)

The teaching language is English, but students may communicate with the instructors and write examination answers in Swedish.

1. Course description and objectives

This is a course about how economic research should be conducted and communicated. The course combines lectures on scientific methods and applied econometrics with lab sessions on applied problems and students' individual quizzes and assignments. The course's focus is on understanding, computing, interpreting and communicating empirical findings. The course objectives are presented in detail in the course syllabus.

After completing the course, students should have the knowledge and skills needed for conducting empirical research on economic questions.

2. Literature and other resources

Wooldridge, Jeffrey M., *Introductory Econometrics: A Modern Approach* (latest edition).

Hansson, Sven Ove, *The Art of Doing Science*, KTH, Stockholm. Compendium.

Additional material will also be used as literature. Information on additional literature will be provided during the lectures and computer labs. Information about the additional material will be made available on Blackboard. The course literature is the main, but not necessarily the only, basis for the examination. To the extent that discussions and additional material during lectures and computer labs are related to the course objectives, these may also be considered for the examination.

Örebro University School of Business provides an up-to-date econometric software package (*Stata*, latest version), free of charge. You are encouraged to prepare for the computer labs by installing *Stata* on your laptop. The license information and installation instructions will be posted on the course homepage on Blackboard as the course starts. For technical support, contact the IT support by sending an email to itsupport@oru.se.

3. Teaching and course content

The course language is English. Teaching takes the form of lectures and computer labs.

Lecture attendance is not compulsory, but it is strongly recommended given that everything presented during the lectures can be subject to examination. The student is responsible for making sure that he or she is updated on discussions and additional materials presented during the lectures.

The course consists of the following parts:

3.1 Lectures

The Art of Doing Science and Writing a Thesis in Economics (Lecture 1)

It is **strongly recommended to attend** this lecture, which presents the aim, the objectives and the content of the **voluntary** self-assessment tests (see below)

Instructor: Daniela Andrén

What is the purpose and scope of econometrics?

Book: Hansson

*The Nature of Econometrics and Economic Data and Regression Analysis with **Cross-Sectional** Data (Lectures 2-3, 5 and 8)*

Instructor: Daniela Andrén

Wooldridge: Chapters 1-4, 6-9 and 19.

*Regression Analysis with **Time Series** Data (Lecture 4)*

Instructor: Magnus Lodefalk

Wooldridge: Chapters 10–12

***Panel Data** Methods (Lecture 6 and 7)*

Instructor: Daniela Andrén

Wooldridge: Chapters 13 and 14.

Interpretation of estimates across statistical models with examples from old exams (Lecture 8)

Instructor: Daniela Andrén

Wooldridge: Chapters 1-9, 13-14.

Instrumental Variables and Two-Stage Least Squares (Lecture 9)

Instructor: Magnus Lodefalk

Wooldridge: Chapter 15

3.2 Computer lab (Lab sessions 1–2) for the Individual report (Examination code: A004)

Instructor: Magnus Lodefalk

The aim of the computer labs is to provide a crash-course into the knowledge and abilities needed for pursuing rigorous applied econometric work, focusing on panel data using Stata. The content includes data management, good coding practices, multivariate panel regression analysis and tests and responses related both to the econometric questions at hand as well as to the methodological issues.

Handouts and articles will be provided on Blackboard during the course.

The computer labs contain **compulsory** elements (see below **Examination code: A001**). We recommend *full* and *active* attendance at all labs, which, from our experience, is necessary for the students to pass the individual report examination.

3.3. Voluntary Quizzes and Self-assessment tests (SATs) for active learning

3.3.1 Quizzes

Several quizzes with multiple-choice questions that cover the course main textbook are provided in **Blackboard** during the entire duration of the course.

- The quizzes are designed to *continuously* support the student's learning from the beginning until the end of the course.
- Each quiz covers the chapter/chapters presented during lectures.
- Each quiz has a fixed deadline.
- Answering the quizzes will not affect directly the grade of the course, i.e., no bonus points, but randomly selected questions from the banks will be part of the exams.

3.3.2 Self-assessment tests via ORU-Survey platform

Two **voluntary** self-assessment tests (**SATs**) facilitate self-assessment of the knowledge in basic concepts that one student must know at the beginning and at the end of the course. In such way, students self-assess their knowledge in basic statistics and regression analysis (in the beginning) and econometrics (one week before the exam) and identify strengths and areas for development.

- Specific instructions and an **individual link** to the SATs will be sent via email to the **students' email address registered in Ladok**. The sender is Örebro University/Örebro universitet.
 - *If you do not receive this email in your Inbox, please **check your Spam folder**.*
- Students who do not submit their answers will receive two automatic reminders for each SAT: one during the course period, and one a few days before the re-examination.
- There will also be time for Q&A during the *Introductory Lecture*.

4. Written examination, 4 credits (Examination code: A005)

The acquired knowledge and understanding, competence and skills, and judgement and approach of the students will be evaluated based on a Written Examination, which is an individual written exam that takes place in WISEflow, a digital platform for online examination. Read more about digital examinations in WISEflow at Örebro university [here](#).

Signing up for exams is compulsory. If you have not signed up, you will not be allowed to take the exam. You must then wait until the next time the exam is offered. Signing up for exams is done only **online** via “Studenttjänster”, where you will also get information on the sign-up period applicable for your particular exam.

4.1 Adapted examination for students with a documented disability

If you have a documented disability you are entitled to have your exam adapted. Contact one of our disability coordinators in good time before your first exam so that arrangements for an adapted exam can be made. To contact a disability coordinator, go to [Studying with disabilities](#).

4.2 To think of during the exam

Allowed material during the exam is a *simple dictionary* in paper format only.

- For students registered in a program at Örebro University: English-Swedish; * Swedish-English (ordbok)
- For international students: *the official language of their home university.

Answers may be written in English or Swedish.

Read the instructions for each question carefully. Always explain all your answers! All figures used to illustrate an answer should be complete and self-explanatory, i.e., all axes and curves in each figure should be clearly denoted and explained, which will affect the mark.

4.3 Re-examination

A re-examination session will be offered for students who do not attend or who receive the grade Fail in the regular examination. An additional re-examination session is offered prior to the fall semester, usually during the second half of August. For the re-examination date, see the course schedule via <https://schema.oru.se/>.

5. Computer labs (Examination code: A001)

As a student, you are required to attend: **two computer labs** to prepare you for the analysis and writing of the individual report.

A student who, for special reasons, cannot attend one of the two sessions should contact Magnus Lodefalk, the instructor for the labs and the grader of the assignment.

6. Individual report, 3.5 credits (Examination code: A004)

A prerequisite to participate in the examination Individual Report is participation in the two computer lab sessions.

In the individual report, you will apply knowledge, skills and approaches that you have acquired in the course lectures and computer labs. You will demonstrate that you are able to make a quantitative analysis of an economic question and critically discuss both the economic and methodological issues at hand, as well as communicate in writing. Further instructions will be given separately, at the first computer lab.

After submission of the report, feedback will be given in communication with each student. Finally, the acquired knowledge and understanding, competence and skills, and judgement and approach of the students will (in addition to the written examination) be evaluated based on the individual report.

7. Grades

Grades used for the Written examination (Examination code: A005) and the Individual report (Examination code: A004) are Fail (U), Pass (G) and Pass with Distinction (VG) and ECTS grades F–A according to the scale in Table 1. For the compulsory elements of the computer labs (Examination code: A001), the mark Participated (DT) is used.

Table 1 Correspondence between Swedish and ECTS grades

Swedish grade	ECTS grade	Definition
VG	A	Excellent: exceptional answers with only minor and insignificant errors, flawless calculations/derivations, complete illustrations and figures, clear and reasonable assumptions, clear reasoning and consistent discussion; clear and correct interpretation and communication of estimated parameters.
VG	B	Very good: above average standard but with a few errors and shortcomings, and lacks details in comparison with answers awarded the grade A.
G	C	Good: generally good result, but with some errors and shortcomings.
G	D	Satisfactory: decent result but with significant errors and shortcomings.
G	E	Sufficient: the result fulfills the minimum criteria.
U	F	Insufficient: considerably more work is needed.

To obtain the final Swedish grade Pass (G), the student must receive a passing grade on all examination components. In addition, the student must have attended all sessions where compulsory attendance is required.

To obtain the final grade Pass with Distinction (VG), the student must in addition receive the grade Pass with Distinction on the Written examination.

To obtain a final ECTS grade of E or better for the course, the student must receive a minimum grade of E both on the Written examination and the Individual report. In addition, the student

must have attended all sessions where compulsory attendance is required. To arrive at an ECTS grade for the whole course, the grades awarded for each examination assignment, grades E–A, are first converted to the numerical values 1–5. An average value is then calculated, considering the number of credits for each examination assignment in relation to the total number of credits for the course. The course grade is thus awarded by means of a weighted average of the examination assignments in the course.

Note that a grading decision may not be appealed against. However, a grading decision for which a notification has been issued may be rectified either through a correction or review. If you wish to request a correction or review, please go to [Grading decisions – reviews and corrections](#) where you can find a link to the web form to fill out. For further information, see the university's local examination regulations (in Swedish).

8. Cheating

No form of cheating whatsoever is allowed. This includes among other things attempts at cheating and helping someone else to cheat. See further information in [English](#) and in [Swedish](#).

9. Course evaluation

There will be a course evaluation at the end of the course. All students are encouraged to give anonymous and constructive comments to help improve the course by both answering the evaluation survey and giving feedback during the course.

10. Contact information

Administrative details related to the registration for the exam and the grades can *only* be answered by the course administrator.

Name	Role	Office	email
Daniela Andrén	Instructor Course coordinator	N4009 Nova House	daniela.andren@oru.se
Magnus Lodefalk	Instructor	N4008 Nova House	magnus.lodefalk@oru.se
Dorotea Åhlgren	Course administrator	N2007 Nova House	dorotea.ahlgren@oru.se